

Test 2 Review

1 Limits

Problem 1. Evaluate the following limit. If it does not exist write DNE.

$$\lim_{(x,y,z) \rightarrow (-1,1,0)} e^{xyz} + x^2 \cos z - 2x \tan^{-1} y$$

Problem 2. Use the Squeeze Theorem to Evaluate the following limit.

$$\lim_{(x,y) \rightarrow (0,0)} x^4 \sin \left(\frac{1}{x^2 + y^2} \right)$$

Problem 3. Evaluate the following limit. If it does not exist write DNE.

$$\lim_{(x,y) \rightarrow (2,2)} \frac{x - y}{\sqrt{x} - \sqrt{y}}$$

Problem 4. Evaluate the following limit. If it does not exist write DNE.

$$\lim_{(x,y) \rightarrow (2,0)} \frac{2xy - 2y}{x^2 + y^2 - 4x + 4}$$

Problem 5. Evaluate the following limit. If it does not exist write DNE.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^6 - 3y^2 x^3}{y^4 + 2x^6}$$

2 Partial Derivatives

Problem 6. Find both first order partial derivatives of the following surface.

1. $f(x, y) = 4^{\cos(xy)} + 2xy^5$

2. $f(x, y) = x^2 \sqrt[6]{2xy}$

3. $z = \frac{6y}{\ln(x+y)}$

4. $f(x, y) = \cos^2(ye^x)$

Problem 7. Find all second order partial

1. $f(x, y) = \frac{x^3}{y^5} + x^2 - y^7$

2. $f(x, y) = x^{\cos y} + e^x$

Problem 8. Consider the surface $f(x, y) = 5x^3y - 2y^2 - 3x + 1$. Find the direction of steepest ascent at $(2, 2, 67)$ and find the slope of steepest ascent.

3 The Chain Rule

Problem 9. Find $\frac{\partial z}{\partial t}$ if $z = \sqrt{x} - \ln xy$, $x = st + t^2$, and $y = \frac{5s}{t}$.

Problem 10. Find $\frac{df}{d\theta}$ evaluated at $\theta = \pi$ where $f(x, y) = e^{xy^2}$, $x = \cos \theta - \sin^2 \theta$ and $y = \tan\left(\frac{\theta}{4}\right)$.

Problem 11. Find an equation which represents $\frac{\partial z}{\partial m}$ where $z = f_1(x, y, z)$, $x = f_2(s)$, $y = f_3(t)$, $z = f_4(s, t)$, $s = f_5(n)$, $t = f_6(m, n)$.

Problem 12. Find $\frac{dy}{dx}$ if $y^3 + x^3 - xy = 0$

4 Directional Derivatives

Problem 13. Find $\mathcal{D}_{\hat{u}}f(1, 1)$ where $f(x, y) = x^3 - 5y + 3xy^2$ and $\hat{u} = \langle \frac{8}{17}, \frac{15}{17} \rangle$

Problem 14. Find the derivative of $f(x, y) = e^{x^2y} + \frac{y}{6x^3}$ in the direction that the angle $\frac{5\pi}{6}$ makes with the x -axis.

Problem 15. Find the derivative of $f(x, y) = e^{x^2y} + \frac{y}{6x^3}$ in the direction of $\vec{w} = 2\hat{i} - 9\hat{j}$.

5 Tangent Planes and Normal Lines

Problem 16. Find the equation both the tangent plane and normal line to the following surfaces at the given points.

1. $z = \frac{1}{2}x^2 - \frac{1}{3}y^3 + \frac{1}{4}x^4; (2, 3, -3)$

2. $z^2 + xyz = xy - 2xz + yz + 1; (1, 2, 1)$

3. $f(x, y) = \sec x + \ln \frac{xy}{\pi}; (\frac{\pi}{3}, 3e, 3)$

6 Relative Extrema

Problem 17. Find and classify all critical values of the following surface.

$$f(x, y) = 7x - 8y + 2xy - x^2 + y^3$$

Problem 18. Find and classify all critical values of the following surface.

$$f(x, y) = (3x + 4x^3)(y^2 + 2y)$$

Problem 19. Find and classify all critical values of the following surface.

$$f(x, y) = x^3 - 3x^2 + 3y^2 + y^3 - 5$$

Problem 20. Find the absolute maximum and minimum of $f(x, y) = 5x^2 - 3y^3 + 4y - x + 1$ bounded by the region $R = \{(x, y) : 0 \leq x \leq 4, 1 \leq y \leq 8\}$

Problem 21. Find the absolute maximum/minimum and where they occur on the surface $f(x, y) = 2x^2 + 3xy + y^2$ bounded by the triangle with end points $(-4, -1), (3, -1), (1, 2)$.

Problem 22. Use Lagrange multipliers to maximize $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint: $x + 2y = 7$.

Problem 23. Use Lagrange multipliers to maximize $f(x, y) = 8x^2 - 2y$ subject to the constraint: $x^2 + y^2 = 1$.